

IS2207 - ELECTRONIC AND PHYSICAL COMPUTING

TUTE – 1

Charge, Current and Voltage

1. If 30 coulombs of charge pass through a conductor in 10 seconds, what is the current?
2. A current of 1.6 amperes flows through a conductor for 1 second. How many electrons pass through the conductor? (Charge of an electron = $1.6 \times 10^{-19} \text{C}$)

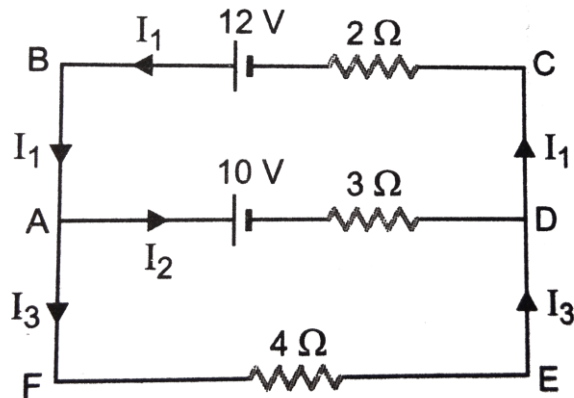
Ohm's Law

1. If a resistor has a resistance of 100Ω and a current of 0.5A flows through it, what is the voltage drop across the resistor?
2. A 220Ω resistor is connected to a 9V battery. Calculate the current flowing through the resistor.
3. An incandescent light bulb draws 0.8A of current when connected to a 120V power source. What is the resistance of the light bulb's filament?
4. A circuit element has a voltage of 3V across it and a current of 15mA flowing through it. Determine the resistance of the element.

Kirchhoff's Laws

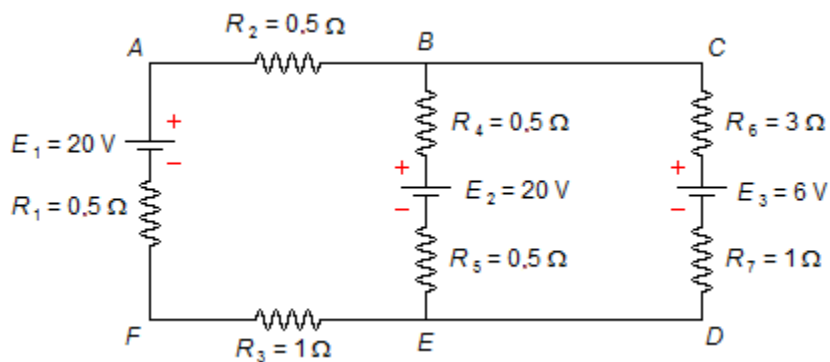
1. Kirchhoff's Current Law (KCL): In a junction, three currents meet. I_1 flows into the junction with a value of 3A , and I_2 flows out of the junction with a value of 1A . What is the value and direction of the third current, I_3 ?
2. Kirchhoff's Voltage Law (KVL): A series circuit consists of a 6V battery and two resistors, $R_1=10\Omega$ and $R_2=20\Omega$. What is the voltage drop across R_1 and R_2 respectively? (Hint: First find the total current).
3. KCL Application: A node in a circuit has four branches connected to it. Currents $I_A=5\text{A}$ and $I_B=2\text{A}$ flow into the node. Current $I_C=7\text{A}$ flows out of the node. Determine the magnitude and direction of the fourth current, I_D .

4.



Using kirchhoff's laws, find the currents I_1 , I_2 and I_3

5.



Using kirchhoff's laws, find the currents of every branch

Resistance of a Conductor ($R=\rho L/A$)

1. Calculate the resistance of a copper wire with a length of 50 meters and a cross-sectional area of $1 \times 10^{-6} \text{ m}^2$. (Resistivity of copper $\rho_{\text{copper}} = 1.68 \times 10^{-8} \Omega \text{ m}$).
2. A wire has a resistance of 10Ω . If its length is doubled and its cross-sectional area is halved, what will be its new resistance?
3. An aluminum wire has a resistance of 2Ω . If its length is 10 meters and its resistivity is $2.82 \times 10^{-8} \Omega \text{ m}$, what is its cross-sectional area?
4. Two wires, A and B, are made of the same material and have the same length. Wire A has a diameter of 2mm, and wire B has a diameter of 4mm. What is the ratio of the resistance

of wire A to wire B (R_A/R_B)?

Parallel and Series Resistors

1. Three resistors, $R_1=5\Omega$, $R_2=10\Omega$, and $R_3=15\Omega$, are connected in series to a 30V power supply.
 - a) Calculate the total equivalent resistance.
 - b) Determine the total current flowing through the circuit.
2. Two resistors, $R_1=6\Omega$ and $R_2=12\Omega$, are connected in parallel across a 24V battery.
 - a) Calculate the total equivalent resistance.
 - b) Determine the current flowing through each resistor.
3. A circuit consists of a 10Ω resistor in series with a parallel combination of a 20Ω and a 30Ω resistor.
 - a) Draw the circuit diagram.
 - b) Calculate the total equivalent resistance of the circuit.
4. You have a collection of 100Ω resistors. How would you combine them (series/parallel) to achieve an equivalent resistance of 25Ω ?

Power ($P=VI$, $P=I^2R$, $P=V^2/R$)

1. A light bulb operates at 120V and draws a current of 0.5A. Calculate the power consumed by the light bulb.
2. A heating element has a resistance of 20Ω and a current of 5A flows through it. Calculate the power dissipated by the heating element.
3. An appliance is rated at 1500W and operates on a 240V supply. What is the current drawn by the appliance?
4. A resistor dissipates 100W of power when connected to a 10V source. What is the resistance of the resistor?
5. Compare the power dissipated by a 10Ω resistor when 2A flows through it versus when it has 20V across it.